

Title: Bank Interest Rate Using Method Overriding

■ Basic Information

Program File: Week7BankMethodOverriding.java

Concepts Used: Inheritance, Method Overriding, Runtime Polymorphism, User Input, Switch Statement

■ Objective / Aim

To develop a Java program demonstrating method overriding by defining a common interest rate in RBI and overriding it in SBI, ICICI, and PNB subclasses.

■ System Requirements

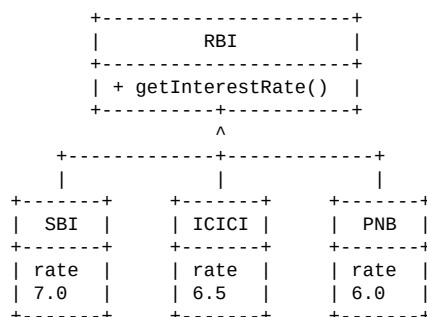
Hardware: Standard PC/Laptop, minimum 4 GB RAM, minimum 500 MB free storage.

Software: Windows/Linux/macOS, Java Development Kit (JDK), Java IDE or Command Prompt/Terminal.

■ Algorithm / Logic Description

1. Start the program.
2. Define RBI with `getInterestRate()` returning 4.0.
3. Define SBI, ICICI, and PNB subclasses overriding the method with their respective rates.
4. Read the bank name, trim spaces, and convert it to uppercase.
5. Use switch to create the matching bank object; unrecognized input defaults to RBI.
6. Display the selected bank's interest rate to one decimal place.
7. Close Scanner and stop.

■ UML Class Diagram



■ Program Flow

```
graph TD
    Start --> EnterBankName[Enter bank name]
    EnterBankName --> TrimInput[Trim and uppercase input]
    TrimInput --> SelectBank[Select bank using switch]
    SelectBank --> DisplayRate[Display interest rate]
    DisplayRate --> CloseScanner[Close Scanner]
    CloseScanner --> End(( ))
```

Start
|
v
Enter bank name
|
v
Trim and uppercase input
|
v
Select bank using switch
|
+--> SBI : 7.0%
+--> ICICI : 6.5%
+--> PNB : 6.0%
+--> Other : RBI 4.0%
|
v
Display interest rate
|
v
Close Scanner
|
v

End

■ Source Code

```
import java.util.Scanner;

class RBI {
    double getInterestRate() { return 4.0; }
}

class SBI extends RBI {
    @Override double getInterestRate() { return 7.0; }
}

class ICICI extends RBI {
    @Override double getInterestRate() { return 6.5; }
}

class PNB extends RBI {
    @Override double getInterestRate() { return 6.0; }
}

public class Week7BankMethodOverriding {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter bank name (RBI/SBI/ICICI/PNB): ");
        String bankName = scanner.nextLine().trim().toUpperCase();
        RBI bank;
        switch (bankName) {
            case "SBI": bank = new SBI(); break;
            case "ICICI": bank = new ICICI(); break;
            case "PNB": bank = new PNB(); break;
            default: bank = new RBI(); bankName = "RBI";
        }
        System.out.printf("%s rate of interest is: %.1f%%\n", bankName, bank.getInterestRate());
        scanner.close();
    }
}
```

■ Compilation & Execution

Step 1: Save the file as **Week7BankMethodOverriding.java**.

Step 2: Compile:

```
javac Week7BankMethodOverriding.java
```

Step 3: Run:

```
java Week7BankMethodOverriding
```

■■ Sample Output

```
Enter bank name (RBI/SBI/ICICI/PNB): SBI  
SBI rate of interest is: 7.0%
```

```
Enter bank name (RBI/SBI/ICICI/PNB): ICICI  
ICICI rate of interest is: 6.5%
```

```
Enter bank name (RBI/SBI/ICICI/PNB): PNB  
PNB rate of interest is: 6.0%
```

```
Enter bank name (RBI/SBI/ICICI/PNB): RBI  
RBI rate of interest is: 4.0%
```

■ Result

The program demonstrates method overriding and runtime polymorphism by displaying the interest rate returned by the selected bank class.